

📝 Employee Management System — Technical Documentation

1. Project Architecture The system utilizes a decoupled **3-Tier Architecture**:

- **Frontend Presentation Layer:** Built with **Angular 19** using modular standalone components, programmatic router navigations, and structural template directives.
- **Backend API Controller Layer:** A lightweight **Node.js Express** server processing incoming requests, enforcing CORS resource access compliance, and coordinating with data repositories.
- **Database Persistence Layer:** A relational **MySQL 8.0** server storing structured rows inside transactional schema layouts.

2. Implemented Features

- Full asynchronous **CRUD Data Management operations** (Create, Read, Update, Delete) synced across layers.
- Interactive data tables featuring responsive hover styling and automatic currency parsing.
- Real-time frontend lookup operations filtering data based on employee name and department parameters.
- Dynamic time-based API query cache-busting logic to guarantee consistent UI-to-database state synchronization.

3. System User Manual

1. Launch the main landing route <http://localhost:4200> to view active workforce rows.
2. Use the "+ Add New Employee" module link to register unique user payloads.
3. Enter alphanumeric strings directly inside the search console field to filter data instantaneously.
4. Trigger localized validation action keys ("Edit" / "Delete") to execute quick row mutations on the fly.